

The Sum, led by Elliott Cisneros, is a Charlottesville, Virginia non-profit partnered with the Heather Heyer Foundation which promotes personal growth, skill development, and diversity. The goal of the The Sum is to stand in solidarity with all people, no matter their background. The Sum offers a Power of Difference Assessment (PDA). The PDA gathers participants' demographics and asks a series of demographic-based questions. After taking the PDA, a report with results is generated and emailed to the participant. The results are categorized across demographics, areas of strength, and areas of growth. The results help reveal people's demographic biases. Those that take the PDA can meet with a consultant from The Sum to learn about their biases and how to communicate better across demographics. There are paid, free, and organizational versions of the PDA with the only difference being the length of the consultation received ("The Sum", n.d.).

The Sum already has an online PDA system in place. The current system allows users to take the PDA and schedule a consultation. However, the system is error-prone. The system improperly categorizes results and it requires someone at The Sum to manually generate reports and email them to users. As part of report generation, categorizations are manually checked and corrected. This makes report generation time consuming and prone to human error. Although manually generating a report only takes a few minutes, the time from PDA completion to reports being emailed to users varies based upon availability at The Sum and can take up to 24 hours. This methodology is not scalable and cannot support the upcoming UVA Department of Psychology study of 1,000 PDA takers. In addition to this, the current system does not detect a difference between assessment versions. It is also insecure and allows for URL manipulation.

The goal of this capstone project is to make a new PDA system. The new system should include all the features of the current system. The new system should correctly categorize results, generate reports, email reports to users, and detect which version of the PDA is being taken. For organizational and paid versions of the PDA, the new system should handle organizational access and payments correctly. The new system should also have security checks in place to prevent revisiting previously answered questions and URL manipulation.

In order to make the new system, requirements had to be gathered from The Sum. Requirements determine what features should be part of the new system and which features should be prioritized. Feature prioritization impacts the development timeline. Requirements help track development progress. Separating the work into requirements allows the team to determine who works on which features. Most importantly, requirements establish clarity between the capstone project team and The Sum for what is to be built.

Minimum requirements are to make a system where users can sign up with a valid email address, undergo email verification, select which version of the assessment to take, fill out user demographics, answer each question of the PDA, only view one question at a time, only answer questions in order, and have access to the separate consultant scheduling system. Users cannot change responses to previously answered questions. For the minimum requirements, assessment versions do not have to differ and The Sum should have administrator access to the system so they can view results, generate reports, and email reports to users.

Desired requirements include having the system correctly categorize results, generate reports, and email the reports to users and The Sum. Desired requirements also include

implementing the paid and organizational versions of the PDA, moving the system to the cloud for scalability, and enhancing the systems administrator experience for The Sum.

Optional requirements include integrating the consultant scheduling system with the PDA system, supporting mobile devices, supporting changing the PDA questions, letting The Sum give consultants permissions to view specific user's results within the system, and allows organizations to view the results for their members who have taken the PDA.

References

The Sum. (n.d.). <http://thesum.org/>.